

Practice Long Quiz 1: Solutions

Logic, Proofs, and Basic Set Theory

Fall 2026

These are full worked solutions to every question on the practice long quiz. They are written the way a complete answer should look on the actual quiz: each step is stated explicitly and justified, definitions are quoted before they are used, and proofs end with a clear concluding sentence. A correct final answer with no justification would *not* receive full credit on the real quiz – use these as a model for how much work to show.

A. Logic – Questions 1–20**40 points**

Question 1. Which of the following are propositions? Briefly justify each answer: (a) “7 is a prime number”; (b) “ $x + 2 = 5$,” where x is an unspecified integer.

Solution. (a) This is a proposition: it is a declarative sentence with a definite truth value (it is true, since 7 has no divisors other than 1 and itself).

(b) This is *not* a proposition: it is an open sentence. Its truth value depends on the value of the unspecified variable x (true if $x = 3$, false otherwise), so it does not have a single, fixed truth value until x is specified.

Question 2. Let p mean “I submit the assignment” and q mean “The assignment is accepted.” Translate into propositional logic: (a) “The assignment is accepted only if I submit it”; (b) “I submit the assignment, but it is not accepted.”

Solution. (a) “ q only if p ” has the form $q \rightarrow p$. So the translation is $q \rightarrow p$.

(b) “But” is logical conjunction, and “it is not accepted” is $\neg q$. So the translation is $p \wedge \neg q$.

Question 3. Suppose p is true, q is false, and r is true. Find the truth value of $(p \wedge \neg q) \rightarrow (q \vee \neg r)$. Show the truth values of the antecedent and the consequent.

Solution. Since q is false, $\neg q$ is true, so the antecedent is $p \wedge \neg q = T \wedge T = T$.

Since r is true, $\neg r$ is false, so the consequent is $q \vee \neg r = F \vee F = F$.

The whole expression is (antecedent) $\rightarrow$ (consequent) $= T \rightarrow F = F$.

So $(p \wedge \neg q) \rightarrow (q \vee \neg r)$ is **false** under this assignment.

Question 4. Write the converse, inverse, and contrapositive of the statement: “If an integer is divisible by 6, then it is even.”

Solution. Let p : “an integer is divisible by 6” and q : “an integer is even.” The statement is $p \rightarrow q$.

Converse ($q \rightarrow p$): “If an integer is even, then it is divisible by 6.”

Inverse ($\neg p \rightarrow \neg q$): “If an integer is not divisible by 6, then it is not even.”

Contrapositive ($\neg q \rightarrow \neg p$): “If an integer is not even, then it is not divisible by 6.”

Question 5. Construct a complete truth table for $p \leftrightarrow \neg q$. Include columns for p , q , $\neg q$, and the final expression.

Solution.

p	q	$\neg q$	$p \leftrightarrow \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	F

Question 6. Use a truth table to determine whether $(p \wedge q) \rightarrow p$ is a tautology, a contradiction, or a contingency.

Solution.

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

The final column is true in every row, so $(p \wedge q) \rightarrow p$ is a **tautology**.

Question 7. Simplify $\neg[p \vee (\neg p \wedge q)]$ using laws of logic. Name a law used at each step. Do not use a truth table.

Solution.

$$\begin{aligned}
 p \vee (\neg p \wedge q) &\equiv (p \vee \neg p) \wedge (p \vee q) && \text{Distributive law} \\
 &\equiv T \wedge (p \vee q) && \text{Negation law} \\
 &\equiv p \vee q && \text{Identity law}
 \end{aligned}$$

Therefore

$$\neg[p \vee (\neg p \wedge q)] \equiv \neg(p \vee q) \equiv \neg p \wedge \neg q \quad (\text{De Morgan's law}).$$

Question 8. Use logical equivalence laws to show that $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$. Name the laws you use.

Solution.

$$\begin{aligned}
 p \rightarrow (q \rightarrow r) &\equiv \neg p \vee (q \rightarrow r) && \text{Conditional law} \\
 &\equiv \neg p \vee (\neg q \vee r) && \text{Conditional law} \\
 &\equiv (\neg p \vee \neg q) \vee r && \text{Associative law} \\
 &\equiv \neg(p \wedge q) \vee r && \text{De Morgan's law} \\
 &\equiv (p \wedge q) \rightarrow r && \text{Conditional law}
 \end{aligned}$$

Question 9. Show that $p \rightarrow q$ and $q \rightarrow p$ are not logically equivalent. Give one truth assignment and evaluate both expressions under it.

Solution. Take $p = F$, $q = T$.

$$p \rightarrow q = F \rightarrow T = T.$$

$$q \rightarrow p = T \rightarrow F = F.$$

Since the two expressions disagree (T versus F) under this assignment, $p \rightarrow q \not\equiv q \rightarrow p$.

Question 10. Rewrite $\neg(p \leftrightarrow q)$ in disjunctive normal form (an OR of ANDs of variables or their negations). Show your equivalence steps.

Solution. Start from the standard expansion of the biconditional:

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q).$$

Negate both sides and apply De Morgan's law twice:

$$\begin{aligned} \neg(p \leftrightarrow q) &\equiv \neg[(p \wedge q) \vee (\neg p \wedge \neg q)] \\ &\equiv \neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q) && \text{De Morgan's law} \\ &\equiv (\neg p \vee \neg q) \wedge (p \vee q) && \text{De Morgan's law (twice)} \end{aligned}$$

Now distribute:

$$\begin{aligned} (\neg p \vee \neg q) \wedge (p \vee q) &\equiv (\neg p \wedge p) \vee (\neg p \wedge q) \vee (\neg q \wedge p) \vee (\neg q \wedge q) && \text{Distributive law} \\ &\equiv F \vee (\neg p \wedge q) \vee (p \wedge \neg q) \vee F && \text{Negation law} \\ &\equiv (p \wedge \neg q) \vee (\neg p \wedge q) && \text{Identity law} \end{aligned}$$

This is an OR of ANDs of literals, so it is in disjunctive normal form:

$$\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q).$$

Question 11. Let S be a set of students, B a set of books, and $R(s, b)$ mean “student s has read book b .” Express “Every student has read at least one book” using quantifiers. Specify the domain of each variable.

Solution.

$$\forall s \in S \exists b \in B R(s, b).$$

Here s ranges over the domain S and b ranges over the domain B .

Question 12. Let D be a set of students, and let $A(x)$ mean “ x is absent.” Express “Exactly one student is absent” using quantifiers, equality, and logical connectives. Do not use the symbol $\exists!$.

Solution. “Exactly one” means: there is a student who is absent, and every other student is not absent.

$$\exists x \in D \left(A(x) \wedge \forall y \in D (y \neq x \rightarrow \neg A(y)) \right).$$

Question 13. Negate $\forall x \in \mathbb{R} \exists y \in \mathbb{R} (x + y = 0)$. Push the negation through every quantifier and express the result using $\neq$, with no negation symbol in front of a quantifier.

Solution.

$$\begin{aligned} \neg[\forall x \exists y (x + y = 0)] &\equiv \exists x \neg[\exists y (x + y = 0)] && \neg\forall \equiv \exists\neg \\ &\equiv \exists x \forall y \neg(x + y = 0) && \neg\exists \equiv \forall\neg \\ &\equiv \exists x \in \mathbb{R} \forall y \in \mathbb{R} (x + y \neq 0). \end{aligned}$$

Question 14. Determine the truth value of each statement and justify your answers: (a) $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (y > x)$; (b) $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z} (y > x)$.

Solution. (a) **True.** Given any integer x , the integer $y = x + 1$ satisfies $y > x$. Since such a y exists for every x , the statement holds.

(b) **False.** Suppose such a y existed with $y > x$ for every $x \in \mathbb{Z}$. Taking $x = y$ gives $y > y$, which is false. So no single y can exceed every integer, and the statement is false.

Question 15. Determine whether $\exists x (x^2 = 2)$ is true when the domain is (a) $\mathbb{Z}$ and (b) $\mathbb{R}$. Justify each answer.

Solution. (a) **False over $\mathbb{Z}$.** If $x^2 = 2$ for an integer x , then x must lie strictly between 1 (since $1^2 = 1 < 2$) and 2 (since $2^2 = 4 > 2$), but no integer lies strictly between 1 and 2. So no integer satisfies $x^2 = 2$.

(b) **True over $\mathbb{R}$.** The real number $x = \sqrt{2}$ satisfies $x^2 = (\sqrt{2})^2 = 2$, so a witness exists.

Question 16. From the premises $p \rightarrow q$, $q \rightarrow r$, and p , derive r . Give a short sequence of statements and name the inference rule used at each step.

Solution.

1. p (premise)
2. $p \rightarrow q$ (premise)
3. q (modus ponens, 1 and 2)
4. $q \rightarrow r$ (premise)
5. r (modus ponens, 3 and 4)

Question 17. Is the argument with premises $p \rightarrow q$ and q , and conclusion p , valid? If valid, justify it; if invalid, give a truth assignment making both premises true and the conclusion false.

Solution. This argument is **invalid** (it is the fallacy of affirming the consequent). Take $p = F$, $q = T$: then $p \rightarrow q = F \rightarrow T = T$ (premise true) and $q = T$ (premise true), but the conclusion $p = F$ is false. Both premises are true while the conclusion is false, so the argument is not valid.

Question 18. All variables range over students. From $\exists x T(x)$ and $\forall x (T(x) \rightarrow P(x))$, derive $\exists x P(x)$. Explain how you use a witness for the existential premise.

Solution. From $\exists x T(x)$, by existential instantiation there is some particular student a (a witness) with $T(a)$ true.

Applying the universal premise $\forall x (T(x) \rightarrow P(x))$ to this particular a (universal instantiation) gives $T(a) \rightarrow P(a)$.

Since $T(a)$ is true, modus ponens gives $P(a)$ true.

Finally, since $P(a)$ holds for this particular a , existential generalization gives $\exists x P(x)$. ■

Question 19. Can $p \vee q$, $\neg p \vee r$, $\neg q$, and $\neg r$ all be true at once? Give a satisfying truth assignment or explain why none exists.

Solution. Suppose all four were true. From $\neg q$ true, $q = F$. From $\neg r$ true, $r = F$. Since $p \vee q$ is true and $q = F$, we need $p = T$. Since $\neg p \vee r$ is true and $p = T$ (so $\neg p = F$), we need $r = T$. But we already established $r = F$ – a contradiction. So **no** truth assignment makes all four true simultaneously.

Question 20. Negate $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} \forall z \in \mathbb{Z} (x + y > z)$. Push the negation through all three quantifiers and express the result using $\leq$.

Solution.

$$\begin{aligned}
 \neg[\forall x \exists y \forall z (x + y > z)] &\equiv \exists x \neg[\exists y \forall z (x + y > z)] \\
 &\equiv \exists x \forall y \neg[\forall z (x + y > z)] \\
 &\equiv \exists x \forall y \exists z \neg(x + y > z) \\
 &\equiv \exists x \in \mathbb{Z} \forall y \in \mathbb{Z} \exists z \in \mathbb{Z} (x + y \leq z).
 \end{aligned}$$

B. Proofs – Questions 21–40**40 points**

Question 21. Give a direct proof that the sum of two odd integers is even. Start from the definition of an odd integer.

Solution. Let m and n be odd integers. By definition of odd, $m = 2a + 1$ and $n = 2b + 1$ for some integers a, b .

Then

$$m + n = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

Since $a + b + 1$ is an integer, $m + n$ is 2 times an integer, so $m + n$ is even by definition. ■

Question 22. Give a direct proof that the product of two rational numbers is rational. State why the denominator in your resulting fraction is nonzero.

Solution. Let r and s be rational numbers. By definition, $r = \frac{a}{b}$ and $s = \frac{c}{d}$ for some integers a, b, c, d with $b \neq 0$ and $d \neq 0$.

Then

$$rs = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Here ac and bd are integers (products of integers), and $bd \neq 0$ because $b \neq 0$ and $d \neq 0$ (the product of two nonzero integers is nonzero). So rs is a ratio of integers with nonzero denominator, i.e., rs is rational. ■

Question 23. Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. Prove directly that if $a \mid b$ and $a \mid c$, then $a \mid (3b - 2c)$. Use the definition of divisibility.

Solution. Suppose $a \mid b$ and $a \mid c$. By definition of divisibility, $b = ak$ and $c = a\ell$ for some integers k, ℓ .

Then

$$3b - 2c = 3(ak) - 2(a\ell) = a(3k - 2\ell).$$

Since $3k - 2\ell$ is an integer, this shows $3b - 2c$ is a times an integer, so $a \mid (3b - 2c)$. ■

Question 24. Prove by contraposition that, for every $n \in \mathbb{Z}$, if n^2 is even, then n is even. State the contrapositive explicitly.

Solution. The contrapositive of “if n^2 is even then n is even” is “if n is odd then n^2 is odd.” We prove the contrapositive.

Suppose n is odd, so $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is of the form $2(\text{integer}) + 1$, so n^2 is odd.

Since the contrapositive is true, the original statement (if n^2 is even, then n is even) is true. ■

Question 25. Let $m, n \in \mathbb{Z}$. Prove by contraposition that if $m + n$ is odd, then at least one of m and n is odd.

Solution. The contrapositive is: if neither m nor n is odd (i.e., both are even), then $m + n$ is even. We prove this contrapositive.

Suppose m and n are both even, so $m = 2a$ and $n = 2b$ for some integers a, b . Then

$$m + n = 2a + 2b = 2(a + b),$$

which is even since $a + b$ is an integer.

Since the contrapositive holds, the original statement holds: if $m + n$ is odd, then at least one of m, n is odd. ■

Question 26. Prove by cases that $3 \mid n(n+1)(n+2)$ for every $n \in \mathbb{Z}$. Use the three cases $n = 3k$, $n = 3k + 1$, and $n = 3k + 2$.

Solution. Every integer n can be written in exactly one of the forms $3k$, $3k + 1$, or $3k + 2$ for some integer k . We check each case.

Case 1: $n = 3k$. Then $n(n+1)(n+2) = 3k(3k+1)(3k+2)$, which has $3k$ as a factor, so it is divisible by 3.

Case 2: $n = 3k + 1$. Then $n + 2 = 3k + 3 = 3(k + 1)$, so $n(n+1)(n+2) = (3k+1)(3k+2) \cdot 3(k+1)$, which has the factor $3(k+1)$, so it is divisible by 3.

Case 3: $n = 3k + 2$. Then $n + 1 = 3k + 3 = 3(k + 1)$, so $n(n+1)(n+2) = (3k+2) \cdot 3(k+1) \cdot (3k+4)$, which has the factor $3(k+1)$, so it is divisible by 3.

In every case, $3 \mid n(n+1)(n+2)$. Since these three cases exhaust all integers, the statement holds for every $n \in \mathbb{Z}$. ■

Question 27. Prove by contradiction that there is no greatest rational number.

Solution. Suppose, for contradiction, that there is a greatest rational number r ; that is, $r \in \mathbb{Q}$ and $r \geq s$ for every $s \in \mathbb{Q}$.

Consider $r + 1$. Since r and 1 are both rational, and the sum of two rationals is rational, $r + 1 \in \mathbb{Q}$. Also, $r + 1 > r$.

But this contradicts the assumption that r is greater than or equal to every rational number, since $r + 1$ is a rational number exceeding r .

This contradiction shows the assumption is false, so there is no greatest rational number. ■

Question 28. Let r be rational and x be irrational. Prove by contradiction that $r + x$ is irrational. You may use the fact that the difference of two rational numbers is rational.

Solution. Suppose, for contradiction, that $r + x$ is rational. Call this rational number s , so $s = r + x$.

Then $x = s - r$. Since s and r are both rational, and the difference of two rational numbers is rational, x would have to be rational.

This contradicts the hypothesis that x is irrational.

Hence the assumption was false, and $r + x$ is irrational. ■

Question 29. Disprove the claim: “For every integer $n \geq 0$, the number $n^2 + n + 41$ is prime.” Give a counterexample and a factorization showing that its value is composite.

Solution. To disprove a universal statement, it suffices to give one counterexample.

Take $n = 40$:

$$n^2 + n + 41 = 40^2 + 40 + 41 = 1600 + 40 + 41 = 1681.$$

Now factor: $41^2 = 1681$, so

$$1681 = 41 \times 41.$$

Since 1681 has 41 as a proper divisor (and $41 \neq 1, 1681$), 1681 is composite, not prime. This contradicts the claim, so the claim is false.

Question 30. Prove that, for every $n \in \mathbb{Z}$, n is even if and only if $3n + 2$ is even. Clearly establish both directions.

Solution. ($\Rightarrow$) Suppose n is even, so $n = 2k$ for some integer k . Then

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1),$$

which is even since $3k + 1$ is an integer.

($\Leftarrow$) Suppose $3n + 2$ is even. We show n is even by proving the contrapositive: if n is odd, then $3n + 2$ is odd. Suppose $n = 2j + 1$ for some integer j . Then

$$3n + 2 = 3(2j + 1) + 2 = 6j + 5 = 2(3j + 2) + 1,$$

which is odd. So if n is odd, $3n + 2$ is odd; equivalently, if $3n + 2$ is even, n is even.

Both directions hold, so n is even if and only if $3n + 2$ is even. ■

Question 31. A student writes: “Let m, n be odd integers. Then $m = 2k + 1$ and $n = 2k + 1$ for some integer k , so $m + n = 4k + 2$ is even.” Explain the unjustified assumption and repair the proof.

Solution. Error. The student uses the *same* integer k for both m and n . This silently assumes $m = n$, which is not given – m and n are two (possibly different) odd integers, so they need two independent parameters.

Repaired proof. Let m, n be odd integers. By definition, $m = 2k + 1$ and $n = 2j + 1$ for some integers k, j (not necessarily equal). Then

$$m + n = (2k + 1) + (2j + 1) = 2k + 2j + 2 = 2(k + j + 1),$$

which is even since $k + j + 1$ is an integer. ■

Question 32. Prove by mathematical induction that $\sum_{i=1}^n (2i - 1) = n^2$ for every integer $n \geq 1$. Include the base case, induction hypothesis, and inductive step.

Solution. Base case ($n = 1$). The left side is $2(1) - 1 = 1$, and the right side is $1^2 = 1$. They are equal, so $P(1)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 1$, i.e.,

$$\sum_{i=1}^k (2i - 1) = k^2.$$

Inductive step. We show $P(k + 1)$ holds, i.e., $\sum_{i=1}^{k+1} (2i - 1) = (k + 1)^2$.

$$\begin{aligned} \sum_{i=1}^{k+1} (2i - 1) &= \sum_{i=1}^k (2i - 1) + (2(k + 1) - 1) \\ &= k^2 + (2k + 1) && \text{by the inductive hypothesis} \\ &= (k + 1)^2. \end{aligned}$$

So $P(k + 1)$ holds.

Since the base case holds and $P(k) \Rightarrow P(k + 1)$ for every $k \geq 1$, by mathematical induction $P(n)$ holds for every integer $n \geq 1$. ■

Question 33. Prove by mathematical induction that $3 \mid (4^n - 1)$ for every integer $n \geq 0$. Include the base case and the inductive step.

Solution. Base case ($n = 0$). $4^0 - 1 = 1 - 1 = 0$, and $3 \mid 0$. So $P(0)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 0$: $3 \mid (4^k - 1)$, i.e., $4^k - 1 = 3m$ for some integer m .

Inductive step. We show $P(k+1)$: $3 \mid (4^{k+1} - 1)$.

$$\begin{aligned}
 4^{k+1} - 1 &= 4 \cdot 4^k - 1 \\
 &= 4(4^k - 1) + 4 - 1 \\
 &= 4(4^k - 1) + 3 \\
 &= 4(3m) + 3 && \text{by the inductive hypothesis} \\
 &= 3(4m + 1).
 \end{aligned}$$

Since $4m + 1$ is an integer, $3 \mid (4^{k+1} - 1)$, so $P(k+1)$ holds.

By mathematical induction, $3 \mid (4^n - 1)$ for every integer $n \geq 0$. ■

Question 34. Complete a strong-induction proof that every integer $n \geq 2$ is a product of primes (a single prime counts). The base case $n = 2$ holds. State the strong induction hypothesis and prove the step for $n = k + 1$ by treating the prime and composite cases.

Solution. **Base case** ($n = 2$). 2 is prime, and a single prime counts as a (trivial) product of primes. So $P(2)$ holds.

Strong induction hypothesis. Assume that for some integer $k \geq 2$, every integer j with $2 \leq j \leq k$ is a product of primes.

Inductive step. We show $k + 1$ is a product of primes. There are two cases.

Case 1: $k + 1$ is prime. Then $k + 1$ is trivially a product of one prime, itself.

Case 2: $k + 1$ is composite. Then, by definition of composite, $k + 1 = ab$ for some integers a, b with $2 \leq a, b \leq k$ (since a proper factorization has both factors strictly between 1 and $k + 1$). By the strong induction hypothesis, a and b are each products of primes:

$$a = p_1 p_2 \cdots p_r, \quad b = q_1 q_2 \cdots q_s.$$

Then

$$k + 1 = ab = p_1 p_2 \cdots p_r q_1 q_2 \cdots q_s,$$

a product of primes.

In both cases, $k + 1$ is a product of primes, so $P(k + 1)$ holds.

By strong induction, every integer $n \geq 2$ is a product of primes. ■

Question 35. A student claims $2^n \geq n + 2$ for all integers $n \geq 0$, using only: “If $2^k \geq k + 2$, then $2^{k+1} \geq 2k + 4 \geq k + 3$.” Explain why this is not a proof of the stated claim. Find the smallest starting value of n for which this step and a valid base case prove it.

Solution. **Why it is not a proof.** An inductive step alone, with no verified base case, proves nothing – it only shows $P(k) \Rightarrow P(k + 1)$; it never establishes that P is true for any starting value at all.

In fact the claim as stated is *false* for small n : at $n = 0$, $2^0 = 1$ but $0 + 2 = 2$, and $1 \geq 2$ is false. At $n = 1$, $2^1 = 2$ but $1 + 2 = 3$, and $2 \geq 3$ is false. So no base case exists at $n = 0$ or $n = 1$, and the induction cannot even start there.

Finding a valid base case. Check $n = 2$: $2^2 = 4$ and $2 + 2 = 4$, so $4 \geq 4$ is true. So $n = 2$ works as a base case.

Checking the given step is valid for $k \geq 2$. From $2^k \geq k + 2$, multiplying both sides by 2 gives $2^{k+1} \geq 2k + 4$. Also $2k + 4 \geq k + 3$ whenever $k \geq -1$, which certainly holds for $k \geq 2$. So $2^{k+1} \geq k + 3 = (k + 1) + 2$, i.e., $P(k + 1)$ holds.

Conclusion. With base case $n = 2$ and the given inductive step (valid for $k \geq 2$), mathematical induction proves $2^n \geq n + 2$ for every integer $n \geq 2$ – *not* for all $n \geq 0$ as originally claimed. The smallest starting value is $n = 2$.

Question 36. Prove by mathematical induction that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for every integer $n \geq 1$. Include the base case, induction hypothesis, and inductive step.

Solution. **Base case** ($n = 1$). The left side is 1, and the right side is $\frac{1(1+1)}{2} = 1$. They are equal, so $P(1)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 1$, i.e., $\sum_{i=1}^k i = \frac{k(k+1)}{2}$.

Inductive step. We show $P(k+1)$:

$$\begin{aligned} \sum_{i=1}^{k+1} i &= \sum_{i=1}^k i + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) && \text{by the inductive hypothesis} \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is exactly $\frac{(k+1)((k+1)+1)}{2}$, so $P(k+1)$ holds.

Since the base case holds and $P(k) \Rightarrow P(k+1)$ for every $k \geq 1$, by mathematical induction $P(n)$ holds for every integer $n \geq 1$. ■

Question 37. Prove by mathematical induction that $2^n > n$ for every integer $n \geq 1$.

Solution. **Base case** ($n = 1$). $2^1 = 2$ and $2 > 1$, so $P(1)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 1$, i.e., $2^k > k$.

Inductive step. We show $P(k+1)$: $2^{k+1} > k+1$.

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k > 2k && \text{by the inductive hypothesis} \\ &\geq k+1 && \text{since } k \geq 1. \end{aligned}$$

So $2^{k+1} > k+1$, and $P(k+1)$ holds.

By mathematical induction, $2^n > n$ for every integer $n \geq 1$. ■

Question 38. Prove by mathematical induction that $6 \mid (n^3 - n)$ for every integer $n \geq 0$. Include the base case and the inductive step.

Solution. **Base case** ($n = 0$). $0^3 - 0 = 0$, and $6 \mid 0$. So $P(0)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 0$: $k^3 - k = 6m$ for some integer m .

Inductive step. We show $P(k+1)$: $6 \mid ((k+1)^3 - (k+1))$.

$$\begin{aligned} (k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3k^2 + 3k \\ &= 6m + 3k(k+1) && \text{by the inductive hypothesis.} \end{aligned}$$

Since k and $k+1$ are consecutive integers, one of them is even, so $k(k+1)$ is even, say $k(k+1) = 2t$ for an integer t . Then

$$(k+1)^3 - (k+1) = 6m + 3(2t) = 6m + 6t = 6(m+t),$$

which is divisible by 6. So $P(k+1)$ holds.

By mathematical induction, $6 \mid (n^3 - n)$ for every integer $n \geq 0$. ■

Question 39. Prove by mathematical induction that a set with n elements has exactly 2^n subsets, for every integer $n \geq 0$. In the inductive step, split the subsets of a set with $k+1$ elements into those that contain the new element and those that do not.

Solution. **Base case** ($n = 0$). The only set with 0 elements is $\emptyset$, and its only subset is $\emptyset$ itself, so it has $1 = 2^0$ subset. $P(0)$ holds.

Inductive hypothesis. Assume $P(k)$ holds for some integer $k \geq 0$: every set with k elements has exactly 2^k subsets.

Inductive step. Let A be a set with $k+1$ elements. Fix one element $x \in A$ and let $A' = A \setminus \{x\}$, so A' has k elements. Every subset of A either contains x or does not:

- Subsets of A not containing x are exactly the subsets of A' . By the inductive hypothesis, there are 2^k of these.
- Subsets of A containing x are exactly the sets of the form $S \cup \{x\}$ where S is a subset of A' . This is a one-to-one correspondence with the subsets of A' , so there are also 2^k of these.

These two collections are disjoint and together account for every subset of A , so the total number of subsets of A is

$$2^k + 2^k = 2 \cdot 2^k = 2^{k+1}.$$

So $P(k+1)$ holds.

By mathematical induction, a set with n elements has exactly 2^n subsets, for every integer $n \geq 0$. ■

Question 40. Use strong induction to prove that every integer $n \geq 2$ can be written as a sum of 2s and 3s (one or more terms, each equal to 2 or 3). State the base cases needed and complete the inductive step for $k+1 \geq 5$.

Solution. **Base cases.** Since the inductive step below will reduce n by 2, we need two consecutive base cases. $n = 2$: $2 = 2$, a sum of one term. $n = 3$: $3 = 3$, a sum of one term. So $P(2)$ and $P(3)$ hold.

Strong induction hypothesis. Assume that for some integer $k \geq 3$, every integer j with $2 \leq j \leq k$ can be written as a sum of 2s and 3s.

Inductive step. We show $k+1$ can be written as a sum of 2s and 3s, for $k+1 \geq 5$ (i.e., $k \geq 4$).

Consider $(k+1) - 2 = k - 1$. Since $k \geq 4$, we have $k - 1 \geq 3 \geq 2$, and also $k - 1 \leq k$, so $k - 1$ is in the range covered by the strong induction hypothesis. Thus $k - 1$ can be written as a sum of 2s and 3s:

$$k - 1 = x_1 + x_2 + \cdots + x_r, \quad \text{each } x_i \in \{2, 3\}.$$

Adding one more term equal to 2 gives

$$k + 1 = (k - 1) + 2 = x_1 + x_2 + \cdots + x_r + 2,$$

which is again a sum of 2s and 3s. So $P(k+1)$ holds.

By strong induction (with base cases $n = 2, 3$), every integer $n \geq 2$ can be written as a sum of 2s and 3s. ■

C. Basic Set Theory – Questions 41–50

20 points

Question 41. Write the set $\{x \in \mathbb{Z} \mid -3 \leq x < 4 \text{ and } x^2 \leq 4\}$ in roster form, and give its cardinality.

Solution. The condition $-3 \leq x < 4$ restricts x to $\{-3, -2, -1, 0, 1, 2, 3\}$.

The condition $x^2 \leq 4$ restricts x to $\{-2, -1, 0, 1, 2\}$ (since $3^2 = 9 > 4$ and $(-3)^2 = 9 > 4$ fail, while $2^2 = 4 \leq 4$ passes).

Intersecting these two conditions:

$$\{x \in \mathbb{Z} \mid -3 \leq x < 4 \text{ and } x^2 \leq 4\} = \{-2, -1, 0, 1, 2\}.$$

Its cardinality is 5.

Question 42. Let $A = \{\emptyset, \{1\}, 1\}$. Determine whether each statement is true or false, with a brief reason: (a) $\emptyset \in A$; (b) $\{1\} \subseteq A$; (c) $\{\emptyset\} \in A$; (d) $\{\{1\}\} \subseteq A$.

Solution. The elements of A are exactly $\emptyset$, $\{1\}$, and 1 .

(a) **True.** $\emptyset$ is literally listed as an element of A .

(b) **True.** $\{1\} \subseteq A$ means every element of $\{1\}$ is in A , i.e., $1 \in A$. Since 1 is an element of A , this holds.

(c) **False.** $\{\emptyset\}$ (the set containing $\emptyset$) is not one of the three listed elements of A ; it is not literally equal to $\emptyset$, $\{1\}$, or 1 .

(d) **True.** $\{\{1\}\} \subseteq A$ means every element of $\{\{1\}\}$ is in A , i.e., $\{1\} \in A$. Since $\{1\}$ is listed as an element of A , this holds.

Question 43. Are $\emptyset$ and $\{\emptyset\}$ equal? Give the cardinality of each set and explain your answer.

Solution. $|\emptyset| = 0$: the empty set has no elements.

$|\{\emptyset\}| = 1$: this set has exactly one element, namely $\emptyset$ itself.

Since the two sets have different cardinalities ($0 \neq 1$), they are **not equal**: $\emptyset \neq \{\emptyset\}$.

Question 44. Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$, and $C = \{3, 2, 1\}$. Determine whether $A \subseteq B$, $A \subsetneq B$, and $A = C$ are true or false. Briefly justify each answer.

Solution. $A \subseteq B$: **True.** Every element of A ($1, 2, 3$) is also an element of B .

$A \subsetneq B$: **True.** $A \subseteq B$ holds, and $A \neq B$ since $4 \in B$ but $4 \notin A$.

$A = C$: **True.** Two sets are equal exactly when they have the same elements, regardless of listed order; A and C both consist of exactly $\{1, 2, 3\}$.

Question 45. List every element of $\mathcal{P}(\{a, b\})$, where $a \neq b$. Use braces carefully to distinguish elements from subsets.

Solution. The subsets of $\{a, b\}$ are: $\emptyset$, $\{a\}$, $\{b\}$, and $\{a, b\}$. So

$$\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

Question 46. A finite set A has exactly 32 subsets. Find $|A|$ and the number of proper subsets of A . Show the counting rule you use.

Solution. A finite set with n elements has exactly 2^n subsets. We are told this equals 32:

$$2^n = 32 = 2^5 \implies n = 5.$$

So $|A| = 5$.

The proper subsets of A are all subsets except A itself, so there are

$$2^5 - 1 = 32 - 1 = 31$$

proper subsets.

Question 47. Let $A = \{1, 2, 4, 6\}$ and $B = \{2, 3, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

Solution. $A \cup B$ collects all elements in A or B (or both):

$$A \cup B = \{1, 2, 3, 4, 6, 7\}.$$

$A \cap B$ collects elements in both A and B :

$$A \cap B = \{2, 6\}.$$

Question 48. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 4, 6\}$, and $B = \{2, 3, 6, 7\}$. Find $A \setminus B$ and A^c .

Solution. $A \setminus B$ is the set of elements in A but not in B . Checking each element of A : $1 \notin B$ (keep), $2 \in B$ (remove), $4 \notin B$ (keep), $6 \in B$ (remove). So

$$A \setminus B = \{1, 4\}.$$

$A^c = U \setminus A$ is everything in U not in A :

$$A^c = \{3, 5, 7, 8\}.$$

Question 49. Let $A = \{1, 2, 4\}$ and $B = \{2, 3, 4, 5\}$. Compute the symmetric difference $A \triangle B = (A \setminus B) \cup (B \setminus A)$. Show the two differences before taking their union.

Solution. $A \setminus B$: elements of A not in B . Checking: $1 \notin B$ (keep), $2 \in B$ (remove), $4 \in B$ (remove). So $A \setminus B = \{1\}$.

$B \setminus A$: elements of B not in A . Checking: $2 \in A$ (remove), $3 \notin A$ (keep), $4 \in A$ (remove), $5 \notin A$ (keep). So $B \setminus A = \{3, 5\}$.

Therefore

$$A \triangle B = \{1\} \cup \{3, 5\} = \{1, 3, 5\}.$$

Question 50. In a group of 40 students, 24 study computer science, 21 study mathematics, and 10 study both. How many study at least one of the two subjects? How many study neither? Show your calculation.

Solution. Let C be the set of students studying computer science and M the set studying mathematics, with $|C| = 24$, $|M| = 21$, $|C \cap M| = 10$.

By the inclusion–exclusion principle:

$$|C \cup M| = |C| + |M| - |C \cap M| = 24 + 21 - 10 = 35.$$

So 35 students study at least one of the two subjects.

Out of 40 total students, the number studying neither is

$$40 - 35 = 5.$$